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When Will the Lights Go Out?

Power outages are an inevitable reality across the United States, but their timing is not random. Severe weather, vandalism, suspicious activity, and equipment failures all contribute to a national grid that is constantly under stress. When the power goes out, the impact is immediate. Food spoils, medical equipment fails, businesses shut down, and students lose power right before a deadline. The ability to anticipate outages even hours in advance could fundamentally change how utility companies respond and how communities prepare.

You are a newly hired Data Analyst for the U.S. Department of Energy's Office of Electricity. Your department has been tasked with understanding outage patterns to better prepare utility companies and emergency responders for what is coming. You have been handed a real dataset, the Event-Related Outage Dataset in America from [Data.gov](https://data.gov). It contains roughly 77,000 records of power outages that occurred across the United States in 2023. Each record tells you when an outage began, where it happened, what caused it, and how long it lasted. The patterns hidden in this data have real consequences for millions of Americans who depend on reliable power every day.

Your job is to build a machine learning model that can predict the time of day a power outage will occur based on factors like location, event type, and seasonal patterns. Before modeling, you will conduct a thorough exploratory analysis of temporal patterns in the data, and engineer a new time-based feature to capture what you find. You will compare three modeling approaches, a baseline Linear Regression, a Random Forest, and an XGBoost model to determine which one performs best. Each model brings a different strength to the table, and part of your job is figuring out which one rises to the challenge. You will evaluate your models using standard regression metrics and present your findings in a reproducible GitHub repository.

If utilities can anticipate when outages are most likely to strike, they can pre-position crews, alert customers, and reduce recovery time. That's the real-world impact of the work you are about to do. At the end of this case study you will deliver a clear explanation of your analytical choices and findings, and perhaps prove that with smarter features, you can outperform the analysts who tackled this dataset before you.

The lights are flickering. Get to work.